

Lessons learned from experiments with four pedal-based exergames

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ARTICLE INFO

Keywords:
Exergames
Gameflow
Engagement
Exercise
Motivation

ABSTRACT

The rise in sedentary lifestyles is a growing concern, contributing to health issues such as premature mortality, cardiovascular diseases, and type 2 diabetes. A key factor in this trend is a lack of motivation to exercise. Exergames – video games designed to promote physical activity – offer a potential solution by making exercise more engaging. This study explores four pedal-based exergames, each differing in genre, concept, and controls, to assess their impact on gameflow, enjoyment, motivation, replayability, and exercise intensity. The results indicate that these games generally support positive gameflow, enjoyment, and high-intensity exercise while motivating some individuals to be more active. The study also highlights key differences between the games and presents seven design recommendations for future pedal-based exergames.

1. Introduction

Sedentary behavior is increasingly prevalent in modern society [1] and is a major risk factor for premature mortality [2]. Prolonged inactivity is linked to cardiovascular diseases and type 2 diabetes [3], with screen-based activities – such as watching television and playing video games – being primary contributors [4]. A significant barrier to physical activity is a lack of motivation [5].

Paradoxically, while video games often contribute to sedentary lifestyles, they can also be used to promote physical activity. Exergames – games that involve movement such as strength, balance, or flexibility exercises – offer an active alternative to passive screen time. These games can improve physical, mental, and social well-being, helping players shift from sedentary habits to more active lifestyles [6]. For instance, Pokémon Go has been shown to increase daily step counts by up to 35 % [7], or about 5.6 additional miles of walking per week [8]. Other reported benefits include spending more time outdoors, weight loss, better sleep [9], and reduced sedentary behavior [10,11].

The Physical Activity Guidelines for Americans recommend that adults engage in at least 150 min of moderate-intensity or 75 min of vigorous-intensity aerobic physical activity weekly for significant health benefits [60]. Moderate intensity requires a heart rate of 50–70 % of the

maximum, while vigorous intensity requires 70–80 %. Such heart rate intensity levels are usually not achieved by exergames like Pokémon Go. However, pedal-based exergames can result in high-intensity exercise. Some pedal-based exergames, like Peloton Lanebreak and Zwift [12], simulate riding a real bike, targeting those who enjoy cycling.

To attract a wider audience beyond traditional cycling enthusiasts, it is crucial to design high-intensity exergames that cater to diverse player preferences. To investigate this, we developed and tested four pedal-based exergames that varied in genre, concept, camera, and control scheme. The primary aim of this study is to explore how different game concepts and genres influence players' gaming and exercise experiences. Specifically, the first research question examines how concept and genre affect gameflow; the second investigates their impact on enjoyment, motivation, and replayability; and the third explores their influence on heart rate and overall exercise experience. The results reveal key similarities and differences across the games and lead to seven design recommendations for future high-intensity exergame development.

The article is organized as follows: **Section 2** summarizes the related work. **Section 3** describes the games of the study, and outlines the research goal, questions, and approach. **Section 4** presents the results of the *meta*-study of the four games. **Section 5** discusses the findings and their validity. **Section 6** concludes the paper with key takeaways and

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<https://doi.org/10.1016/j.entcom.2025.101018>

Received 13 February 2025; Received in revised form 18 June 2025; Accepted 7 September 2025

Available online 8 September 2025

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recommendations.

2. Related work

This section begins with a brief overview of exergames and the concept of exergaming. It then presents a selection of relevant works, including full-body exergames, researcher-developed exerbike games, and commercially available exerbike games.

2.1. Introduction to exergaming

Video games are often associated with sedentary lifestyles that can lead to negative health effects. However, exergames counteract this by combining exercise with video gaming [13]. Exergaming involves playing video games that require physical exertion or movement beyond sedentary activities, including strength, balance, and flexibility exercises [14]. Exergames vary in genre, controls, physical exertion required, and gameplay.

2.2. Full-body exergames

A prominent example of an exergame is *Dance Dance Revolution* (DDR), a rhythm-based game launched by Konami in Japan in 1998. DDR encourages regular exercise, provides endurance training, and aids in weight loss [15]. It also benefits individuals with Huntington's disease by motivating them to exercise [16] and it enhances cognitive function and inhibitory control [17].

Another commercially successful exergame is *Wii Fit*, released by Nintendo in 2007. *Wii Fit* uses movement sensors in the controller to support a variety of physical activities such as balance, strength, and aerobics training. Numerous studies have shown its positive effects on balance and strength in older adults [18], sensory weighting and dynamic balance in young adults [19], aerobic activities across different age groups [20], and the overall quality of life for elders [21].

Pokémon Go, released by Niantic in 2016, is arguably the most successful exergame to date due to its massive popularity and impact on players. A systematic literature review revealed that the game significantly enhances physical, mental, and social health as long as players remain active [6]. Furthermore, it has been particularly effective for less physically active individuals and gamers, a demographic often difficult to motivate toward social and physical activity [22].

Exermon, a game similar to *Pokémon Go*, encourages strength training by having players evolve creatures through various body-weight exercises monitored by proximity and accelerometer sensors on smartphones [23]. An evaluation of *Exermon* showed that 50 % of players significantly increased their amount of strength exercises while playing the game.

Ring Fit Adventure, released by Nintendo in 2019, is a successful exergame. This RPG lets players explore, battle enemies, and overcome obstacles using strength and aerobic exercises. Studies show that playing for 30 min, three times a week, maintains or improves fitness [24]. Additionally, playing for 45 min, twice a week for eight weeks, significantly enhances body composition and fitness in adult women [25]. The game also alleviates lower back and buttock pain while boosting pain self-efficacy [26].

Beat Saber, a popular VR rhythm exergame released in 2019, involves slicing incoming cubes with lightsabers in time with music. Players can choose from a growing collection of songs, each with five difficulty levels, and play in campaign, solo, online, or party modes. Studies show that *Beat Saber* provides aerobic exercise comparable to traditional methods but is more motivating [27,28]. Its multi-angle modes further increase physical activity and enjoyment, and improve cognition [29]. The game also positively affects the speed and stability of the "hand-to-mouth" movement for people with multiple sclerosis [30].

2.3. Research-developed exerbike games

Greedy Rabbit is a 2D maze mobile game where a rabbit navigates a maze to reach a carrot, avoiding hedgehogs and collecting flowers along the way [31]. The game connects to an exercise bike via Bluetooth, with the rabbit's speed tied to the cycling speed. The bike's resistance adjusts based on game events. In a study with 163 students, those who played *Greedy Rabbit* showed enhanced situational interest and achieved a good level of physical activity compared to a control group that cycled without the game.

PaperDude is a VR exercise bike game inspired by the arcade game *Paperboy* [32]. Players complete a newspaper route in a suburban neighborhood, scoring points for delivering mail to mailboxes and losing points for crashing into obstacles. The biking speed controls the virtual character's speed, and a Microsoft Kinect camera tracks arm movements for throwing newspapers.

Play'n Ride is a bike-based exergame designed to promote physical activity [33]. Using a sensor on the back wheel of an ordinary bike, players control a flying character's height by their pedaling speed in an endless runner game. It features two themed universes: "space" and "fantasy garden." Tested on 178 participants during the Tour de France, the game was found to be easy and exciting to play.

The *ExerLearn Bike System* is an educational and physical game system for children on an exercise bike [34]. It features two games, Memory Game and ExerMath, and uses sensors to detect forward and backward motion and a push button. In *ExerMath*, players answer questions by pedaling the correct number of cycles (e.g., for $1 + 3$, they pedal 4 cycles). The Memory Game requires players to cycle through items to reveal a picture. Evaluated by eight children, the game raised their heart rates to 60 % of the maximum and was rated favorably.

A study by Shaw et al. investigated the effects of competition (against a ghost) and cooperation (with a virtual trainer) on exercise performance in a VR cycle exergame [35]. Players cycle on a procedurally generated track, avoiding obstacles and collecting bonuses, with speed determined by pedaling pace. A 3D camera tracks movement for steering. Results showed that competitive modes led to greater distances traveled and were more enjoyable than cooperative modes. Another study found a small but significant increase in exercise performance and motivation when using the exergame compared to exercising without it [36]. VR also slightly increased motivation without affecting performance.

Monedero et al. conducted a study comparing the intensity and enjoyment of conventional cycling with an exercise-bike game [37]. Thirty-four healthy participants alternated between conventional stationary cycling and interactive cycling using a GameBike with a Sony PlayStation 2. The game, *Road Fury 2*, required players to pedal to move a vehicle and use the handlebars to steer, providing visual and audio feedback. Results showed that the exergame led to significantly higher exercise intensity and was more enjoyable than conventional cycling.

Life is a Village is a multiplayer exerbike game set in a virtual world [38]. Players control an avatar's speed by pedaling and steer using a wireless PS2 controller. The bike's resistance mimics virtual terrain, increasing uphill and decreasing downhill. Players gather resources to build a village, with some resources harder to obtain. In cooperative mode, two players control the same avatar: one cycles and navigates, while the other fends off enemies with a *Wii Remote*. The game's evaluation was not included in the paper.

2.4. Commercial exerbike games

Zwift, released in 2014, is a popular online cycling game that allows players to train and compete in various virtual worlds. It is seen as valuable exercise, with a pro-athlete calling it "a very good training alternative" [12]. Users appreciate the social aspect, enjoying rides with friends and making connections through the app. *Zwift* simulates realistic road cycling without the dangers of traffic. Studies suggest it is a

viable substitute for real-life cycling events, enhanced by quality graphics, gamification elements, and social interaction [39].

Peloton Lanebreak, launched in 2022, is an exercise bike game where players collect points by riding a wheel on a six-lane track, with bike resistance determining the lane [40]. It features three challenge types: Beats, requiring a cadence over 20 in the correct lane; Breaker, charging the wheel by increasing cadence; and Streams, maintaining cadence within a specific range. The game offers playlists from 5 to 30 min and four difficulty levels, which set the resistance for the lanes.

2.5. Summary

Table 1 summarizes the research and commercial exergames discussed in the related work section. The majority are exerbike games, which vary in both game control mechanisms and genre. All of these games incorporate sensors that detect pedaling, typically used to control forward motion within the game. Some titles also integrate additional input methods such as Kinect, 3D cameras, buttons, handlebars, or PS2 controllers. Most of the exerbike games fall within the simulation genre, focusing on cycling or racing. Experimental results report increased enjoyment and motivation, improved aerobic exercise outcomes, and elevated heart rates.

The remaining exergames employ a wider range of control methods, including dance pads, balance boards, walking, motion sensors, VR controllers, and fitness equipment. These games span multiple genres such as rhythm, role-playing (RPG), sports, and augmented reality (AR) with location-aware features. Reported outcomes include enhanced motivation, improved cognitive function, increased fitness, pain reduction, and overall health benefits.

3. Material and methods

This section presents the exercise platform used in the study, the theoretical foundation for game design, the games analyzed, and the research objectives, questions, and methods.

3.1. The exergame platform – PlayPulse

PlayPulse is an innovative exergame platform designed for exercise bikes, aiming to deliver a comprehensive gaming and physical exercise experience. It originated as a master's project at the Norwegian University of Science and Technology in 2014. Founders Stian Weie and Kristoffer Hagen sought to create an immersive game that also provided an efficient physical workout [41]. Their efforts resulted in a hardware prototype featuring an exercise bike equipped with buttons on the steering handles and a movement sensor on the wheel, connected to a PC. The first game, *Pedal Tanks*, was a third-person multiplayer tank game [42].

Table 1
Summary of Related Work.

Game title	Genre	Exercise	Game controls	Research Results
Dance Dance Revolution	Rhythm	Aerobic	Dance pad	+Motivation, +Aerobic, +Weight loss, +Cognitive
Wii Fit	Sport sim	Aerobic, balance, strength	Balance board	+Balance, +Strength, +Aerobic, +Quality of life
Pokémon Go	Location-aware, AR	Aerobic	Walking	+Physical, +Mental, and + Social health, +Motivation
Exermon	RPG	Strength	Mobile sensors	+Motivation, +Strength
Ring Fit Adventure	RPG	Aerobic, strength	Fitness ring, sensors	+Body composition, +Fitness, +Reduce pain, +Self-efficacy
Beat Saber	VR Rhythm	Aerobic	VR controllers	+Motivation, +Aerobic, +Coordination
Greedy Rabbit	2D maze	Aerobic, strength	Exercise bike	+Aerobic, +Motivation
PaperDude	VR simulation	Aerobic	Exercise bike, Kinect	– No experimental results –
Play'n Ride	Endless runner	Aerobic	Exercise bike	+Enjoyment
ExerLearn	Edutainment	Aerobic	Exercise bike, push button	+Heart rate 60 % of max
Shaw VR cycling	VR simulation	Aerobic	Exercise bike, 3D camera	+Aerobic, +Motivation
RoadFury 2	Simulation	Aerobic	Exercise bike, handlebars	+Enjoyment, +Heart rate
Life is a Village	Simulation	Aerobic	Exercise bike, PS2 controller	– No experimental results –
Zwift	Simulation	Aerobic	Sensor on bike	+Substitute real cycling
Peloton Lanebreak	Lane Racer	Aerobic	Exercise bike	– No experimental results –

Unlike most exergames, PlayPulse offers a richer gaming experience, comparable to popular entertainment games, and includes features from well-known genres like multiplayer online battle arena (MOBA) games [43]. Initial experiments indicated that the game was highly enjoyable and effectively elevated players' heart rates [44]. Subsequent controlled studies demonstrated that *Pedal Tanks* could deliver an enjoyable high-intensity interval training (HIT) workout [45]. One study explored the health benefits of playing PlayPulse for at least 45 min twice a week over eight weeks, revealing a significant increase in peak oxygen uptake [46].

Another study showed that playing PlayPulse in multiplayer mode offered no significant benefits in exercise intensity or perceptual response over single-player mode [47]. However, the higher peak heart rates and a tendency for higher peak VO_2 intensity during multiplayer exergaming imply potential advantages. Finally, a study suggests that the *Pedal Tanks* game can elicit moderate-to-vigorous intensities, making it a viable alternative to conventional aerobic exercise [48].

Today, PlayPulse has evolved into a commercial platform with the PlayPulse One, a fully integrated exercise gaming bike. Additionally, several PlayPulse games have been developed for research to explore various concepts of pedal games.

Fig. 1 illustrates one of the PlayPulse prototype bikes used in the experiments described. The left image shows the bike, including the game controller on the steering handles, the screen, the wheel with the movement sensor, and the PlayPulse control box (the PC connecting the screen and control box is not shown). The middle image features the PlayPulse control box and the left and right thumb buttons, while the right image displays the index and middle buttons for the left hand.

3.2. Game design for player enjoyment

The games discussed in this article were designed using several frameworks to enhance player enjoyment. Malone's theory on intrinsic motivation, originally for learning, can also be applied to exergames. This theory suggests that fun arises from three elements: appropriate challenges, intrinsic fantasy tied to player skill, and sensory and cognitive curiosity [49].

The gameflow model outlines eight elements for player enjoyment: concentration, sufficient challenge, player skill development, player control, clear goals, appropriate feedback, immersion and social interaction [50]. While not all eight elements need to be supported, each can contribute to player enjoyment.

Reward systems are another crucial aspect of game design. Wang and Chuen-Tsai identify eight forms of rewards that are also relevant to exergames: score systems, experience points (XP), item granting, resources, achievements, feedback messages, plot animation and pictures, and unlocking mechanisms [51].

Finally, the Dual Flow model was employed to optimize both exercise effectiveness and game attractiveness, motivating players to



Fig. 1. Pictures of the PlayPulse Prototype Bike.

continue playing and achieve a good workout [52]. Additionally, Malone's theory on intrinsic motivation and the gameflow model were utilized to develop statements in the questionnaires used to evaluate the games discussed in this article.

3.3. The pedal games

The aim of this study was to analyze four pedal games with distinct characteristics, controls, and gameplay mechanics as shown in Fig. 2.

Lane Rider draws inspiration from games like *Temple Run* [53] and *Subway Surfers* [54]. The objective is to navigate a car through an obstacle-filled landscape as quickly as possible (Fig. 2, upper left) [55]. Players control the car by switching between three lanes, jumping, and activating pickups. Obstacles include both static and dynamic elements. Static obstacles, such as large rocks, require lane changes, while smaller objects, like flat stones or coffins, can be jumped over. Dynamic obstacles, including rolling barrels, food trucks, fences, and gates, add an additional challenge. The car's speed is influenced by the player's pedaling rate and track elevation, with uphill sections demanding greater pedaling effort and downhill sections requiring less. Camera movements simulate elevation changes and acceleration to enhance

immersion. Players can collect items that impact gameplay, such as speed boosts and time multipliers, while avoiding penalties like speed reductions and time deductions. Lane Rider is a competitive two-player game where participants compete for the shortest round and total times. A summary of its features is provided in Table 2.

2D Boss Fighter, illustrated in Fig. 2 (upper right) [56], draws inspiration from the game *Cuphead* [57]. In *Cuphead* one or two players control the characters Cuphead and Mugman as they battle through a series of levels and boss fights. Similarly, *2D Boss Fighter* features cooperative gameplay in which two players work together to defeat multiple bosses by reducing their health to zero. The bosses employ a variety of offensive attacks, including projectiles and other combat mechanics. If a player's avatar loses all its health, it dies. However, in two-player mode, the remaining player can revive their teammate by pedaling rapidly to fill a progress bar. Pedaling speed directly influences the shooting rate. Players can switch between two modes: *ground mode*, where the character runs and shoots, and *flying mode*, where a helicopter-like backpack enables aerial movement and attacks. The game features various weapons, including a rocket launcher that requires pedaling to charge before firing. A summary of its characteristics is shown in Table 2.

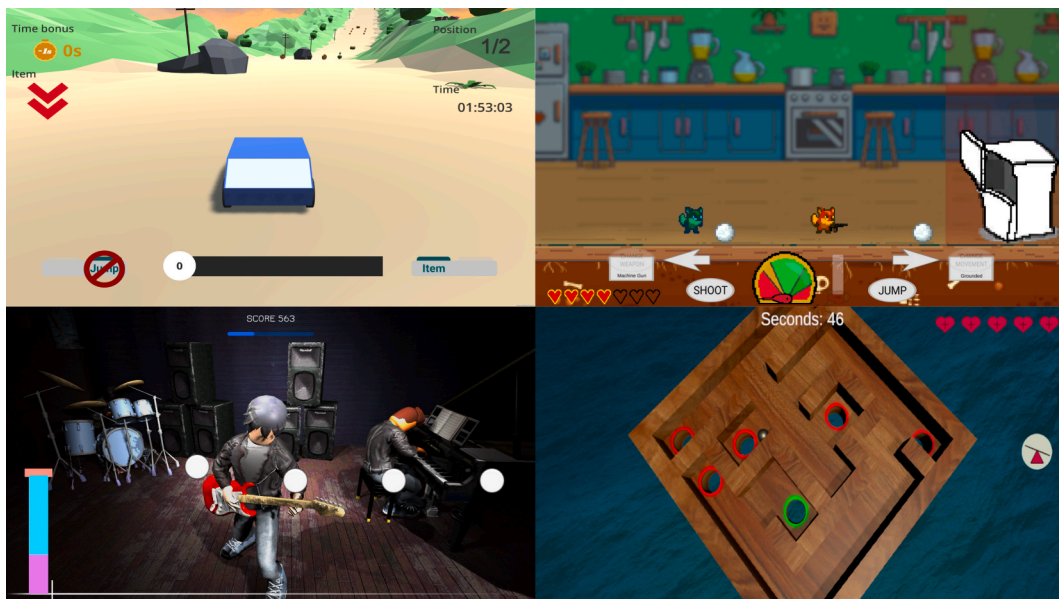


Fig. 2. Representative screenshots from four games: *Lane Rider* (top left), *2D Boss Fighter* (top right), *WaveRider* (bottom left), and *FlipBoard* (bottom right).

Table 2
Summary of Characteristics of the Four Games.

Characteristics	Lane Rider [55]	2D Boss Fighter [56]	WaveRider [61]	FlipBoard [62]
Genre	Endless runner	2D Boss Fighter	Rhythm game	Balance
Inspiration games or toys	Subway Surfers, Temple Run	Cuphead	SingStar, DDR, Guitar Hero	Brio labyrinth, Shape sorting
Multiplayer	2p PvP	1p & 2p PvE	1p, 2p PvP & PvE	1p (scoreboard)
Win state	Quickest race time	Beat the boss	Highest score	Highest score
Camera	3rd person view	2D side static	2D side over 3D scene	3D above and tilted
Pedal	Forward speed	Ground: Shoot speed Flying: Rotor speed Rocket: Launch meter	Height of tone indicator	Rotation angle of board (equilibrium cadence)
Levels	3 tracks	3 boss levels	5 songs	3 levels (balance, shapes, marble)

WaveRider, inspired by music and rhythm games such as *SingStar* [58], *Dance Dance Revolution (DDR)* [59], and *Guitar Hero* [60], is depicted in Fig. 2 (lower right) [61]. The objective is to match the pitch of a song by adjusting pedaling speed and pressing buttons in sync with quick-time events. Pedaling speed controls the height of a visual bar that must align with incoming notes or prompts. The game offers both solo and two-player modes, including *competitive (PvP)* and *cooperative (PvE)* play. In competitive mode (*PvP*), each player manages their own pedaling intensity bar, with the highest score determining the winner. In cooperative mode (*PvE*), both players share a single bar, requiring synchronized effort. The prototype used in this study featured five dynamic music tracks, which increased in intensity as players performed better. A summary of the game’s features is presented in Table 2.

FlipBoard is a puzzle exergame inspired by the *Brio labyrinth* and *shape-sorting* physical toys (Fig. 2, lower right) [62]. Players control a tilting board by adjusting pedaling intensity and using buttons to rotate it. Calibration requires pedaling to fill a loading bar, establishing the equilibrium pedal speed for gameplay. The game features three levels: **Balance** – Players aim to keep objects on the board for as long as possible; **Shapes** – Players earn bonus points by guiding objects through corresponding holes; and **Marble** – The objective is to navigate a metal ball through a maze while avoiding holes that add a 20-second penalty if the ball falls through. Although *FlipBoard* is a single-player game, it features a scoreboard for competitive play. A summary of its characteristics is provided in Table 2.

All the described games are controlled using pedaling and buttons located on the exercise bike’s handlebars. Fig. 3 provides an overview of the control schemes.

3.4. Research questions and research approach

The goal of this meta-study was to investigate the effectiveness of various game genres and concepts on a pedal-based exergame platform. Specifically, we aimed to assess how different pedal games influence player experience both as a game and as a form of exercise. The research method employed was based on the Goal, Question, Metrics (GQM) approach [63]. This involved defining a research goal at a conceptual level, formulating a set of research questions at an operational level, and specifying metrics to answer these questions at a quantitative level. Data was collected from heart rate monitors, surveys, interviews, and observations across four experiments.

3.4.1. Research goal and research questions

Using the GQM template, the research goal was defined as follows:
The purpose of this study was to investigate how game concepts and genres affects players’ game and exercise experience from the point of view of a researcher in the context of exergames on an exercise bike.
The research goal was decomposed into the following three research questions (RQs):

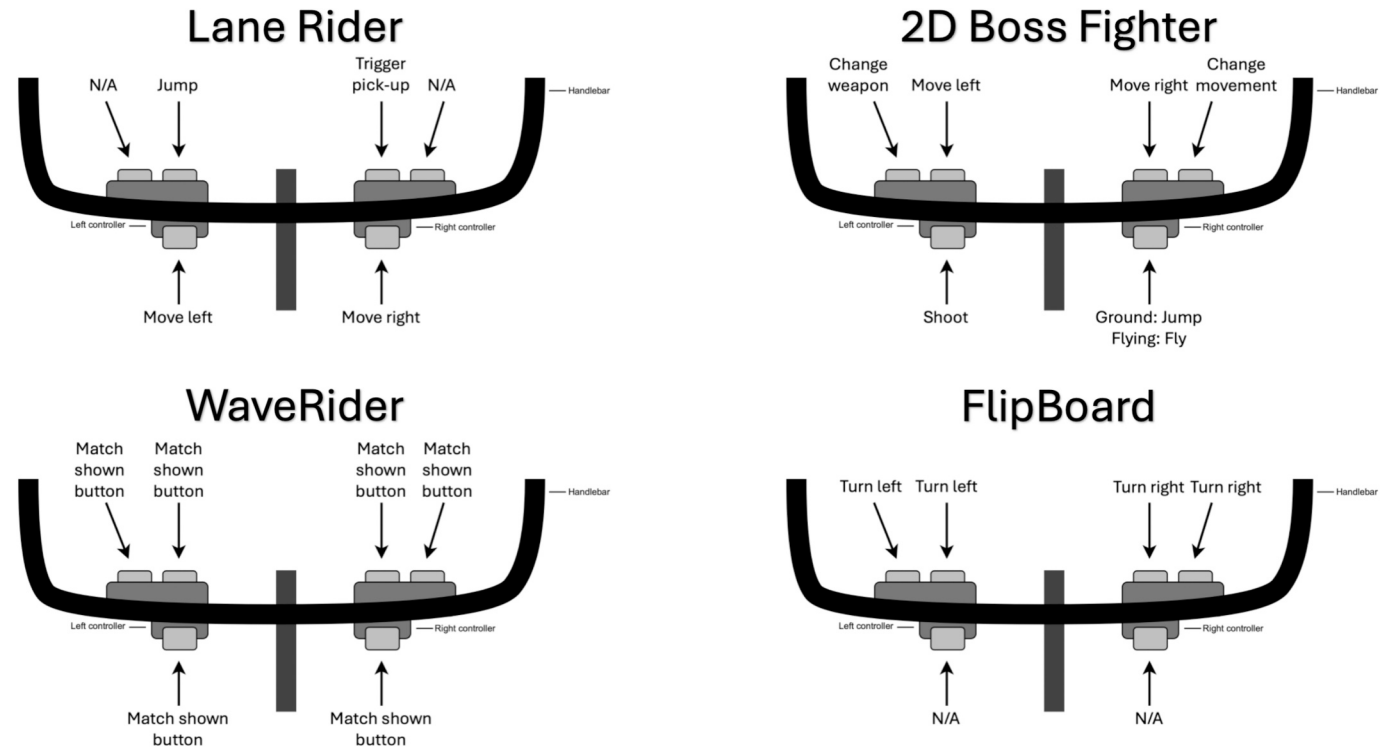


Fig. 3. Overview of the Controls for the Four Pedal Games.

- **RQ1: How do the game concept and genre influence gameflow?** This question explores how different game concepts and genres affect key gameflow elements, including concentration, challenge, player skill, control, clear goals, feedback, immersion, and social interaction.
- **RQ2: How do the game concept and genre impact enjoyment, motivation, and replayability?** This question examines how game design influences player enjoyment, retention (motivation to keep playing), and motivation to engage in physical exercise.
- **RQ3: How do the game concept and genre affect heart rate and exercise experience?** This question investigates how different game concepts and genres influence players' heart rates and overall exercise experiences.

3.5. Data sources and metrics

To address the research questions, data was collected through questionnaires, interviews, observations, and heart rate monitors. The experiments for the four games were conducted over four years (2020–2023) in a lab setting, with each game tested on the PlayPulse exergame platform (see Fig. 1). Participants, typically in pairs, played each game in 20–30 min sessions over multiple days, with some variations in session count and questionnaire usage. Heart rate data was collected using the *Polar H10* monitor in three experiments, while the *Fitbit Charge 3* was used for *Lane Rider*. Table 3 provides a detailed overview of the experimental procedures, including variations in participant demographics, exercise routines, and gaming habits for each game. For *FlipBoard*, exercise and gaming hours per week were estimated based on the number of sessions.

3.6. Data analysis

Descriptive statistics were used to compare and analyze questionnaire results, particularly those employing Likert scales (1 to 5). Mean scores and standard deviations were computed. For statistical significance testing, the Mann-Whitney test was used for two samples [64], while the Kruskal-Wallis test was used for more than two samples [65].

Table 3
Overview of the Four Experiments.

Characteristics	Lane Rider	2D Boss Fighter	WaveRider	FlipBoard
Subjects (N)	16	13	14	10
Age	25.5	23.5	23.5	24.4
Gender	25 % female, 75 % male	15 % female, 85 % male	43 % female, 57 % male	40 % female, 60 % male
Enjoy gaming	50 %	77 %	60 %	60 %
Enjoy exercise	56 %	69 %	65 %	80 %
Exercise/week	4.2 h	5.2 h	2.6 h	2.1 h (estimate)
Gaming/week	8.6 h	7.8 h	8.4 h	7.4 h (estimate)
Sessions	3 (30 min session)	3 (30 min session)	4 (20 min session)	3 (30 min session)
Questionnaire	Post-form	Session forms	Pre-form, Session forms, Post- form	Session forms
Observation	Emotions, verbal interactions, play, experience	Emotional, verbal interactions, play, experience	Emotional, verbal interactions, play, experience	Emotional, verbal interactions, play, experience
Interview	Informal questions after observation	Semi- structured interviews after sessions	Informal questions between games	Formal using interview guide
HR monitor	Fitbit Charge 3	Polar H10	Polar H10	Polar H10

Heart rate data was analyzed by comparing average, standard deviation, maximum and minimum values, median, and 80 % percentiles. Qualitative insights from interviews and observations were used to contextualize and enrich the interpretation of the quantitative data. Any variations in the questionnaires across the four experiments are clearly identified in the results section. However, this article does not report detailed findings from the interviews and observations.

4. Results

This section presents the results of the four game experiments.

4.1. RQ1: How game concept and genre affect gameflow?

The first research question examines how variations in game concept and genre influence gameflow and its eight core elements. These elements are essential for fostering player enjoyment but can be difficult to maintain in exergames, particularly when the physical demands of exercise interfere with gameplay immersion. The results related to gameflow are presented in Table 4, using a Likert scale ranging from Fully Disagree (1) to Fully Agree (5).

Table 4 shows that most games positively influenced the eight elements of gameflow, though there are variations:

- **Concentration:** Most players (average 4.50 +) stated that they were fully focused on the game. There was no statistically significant difference between the games. Observations confirmed the survey results, showing that players were highly concentrated, often forgetting to blink.
- **Challenge:** Only *Lane Rider* and *2D Boss Fighter* included a statement regarding the balance of challenge and skills. Both games provided an appropriate challenge (average 4.00 +). *WaveRider* and *FlipBoard* statements focused on physical challenge, showing that *FlipBoard* offered a more appropriate physical challenge (average 4.56) compared to *WaveRider* (average 3.54). Observations indicated that the challenge largely depended on how much players pushed themselves physically.
- **Player Skills:** Most players felt they improved with each round. Significant differences were found ($p = 0.0009$), with *WaveRider* and *FlipBoard* players feeling a greater sense of skill development compared to *Lane Rider*. This may be due to the more conventional controls in *Lane Rider*.
- **Control:** Players' sense of control varied. *Lane Rider* and *WaveRider* players felt more in control than those playing *2D Boss Fighter*, though the difference was not statistically significant ($p = 0.3166$). The *FlipBoard* survey did not include this statement but had a similar statement about the ease of understanding controls (average 4.00).
- **Clear Goals:** Most players (average above 4.31) understood the game objectives. This consistency was expected due to the simplicity and brevity of the games.
- **Feedback:** Only *2D Boss Fighter* and *FlipBoard* included feedback statements, indicating room for improvement (average 3.42 and 3.22, respectively). Various feedback mechanisms were used in each game, such as position, time, speed, power-ups, lives, and performance indicators. *WaveRider* provided audio feedback, where more and more musical elements were added for successful play.
- **Immersion:** Players generally felt immersed (average of 3.67 +). *WaveRider* provided significantly higher immersion (4.78) compared to *Lane Rider* and *2D Boss Fighter* ($p = 0.0211$), possibly due to the use of music directly linked to gameplay performance [66].
- **Social Interaction:** Most players (average of 3.92 +) enjoyed the social aspects of the games. *FlipBoard* was single-player, apart from a shared scoreboard rated highly (average 5.00). *WaveRider's* competitive mode was preferred by 69 % of players.

Table 4

Results on How Games Affected the Gameflow Scale: 1: Fully Disagree, 2: Disagree, 3: Neutral, 4: Agree, 5: Fully Agree.

Gameflow element	Lane Rider		2D Boss Fighter		WaveRider		FlipBoard	
	Avr	Std	Avr	Std	Avr	Std	Avr	Std
Concentration: Fully focused on game	4.60	0.47	4.50	0.52	4.69	0.39	NA	NA
Challenge: Appropriate skill challenge balance	4.27	0.76	4.00	0.60	NA	NA	NA	NA
Player Skills: Get better as game progresses	3.67	0.61	4.08	0.87	4.85	0.29	4.67	0.46
Control: I had control over situation/game	4.00	0.62	3.33	1.13	3.85	0.74	NA	NA
Clear goals: Understood what to do in the game	4.73	0.43	4.42	0.52	4.31	0.51	4.78	0.35
Feedback: Received useful feedback while playing	NA	NA	3.42	0.93	NA	NA	3.22	1.19
Immersion: Lost sense of time and space	3.67	1.12	3.92	0.63	4.69	0.39	NA	NA
Social Interaction: Enjoyed playing with other players	4.67	0.43	4.67	0.47	3.92	0.94	NA	NA

4.2. RQ2: How game concept and genre affect the enjoyment, motivation and replayability?

Table 5 presents the results related to enjoyment, motivation, and replayability.

Table 5 shows mostly positive results related to enjoyment, motivation and replayability, though there are some variations.

- **Enjoyment:** Most players (average of 4.08 +) enjoyed the games, with no statistically significant differences. Players of *WaveRider* enjoyed competitive gameplay over single-player and collaborative modes. Further, *Lane Rider* players said they enjoyed playing this type of exergame (average 4.47).
- **Motivation:** The games moderately increased players' motivation to exercise (average 3.53 +), with no significant difference ($p = 0.7298$). The *FlipBoard* survey did not include this statement, but players expressed a high desire to exercise by playing the game again (average 4.67).
- **Replayability:** Most players were positive about replaying the games (average 3.77 to 4.60), though *Lane Rider* and *FlipBoard* were slightly more favored ($p = 0.1097$). Players felt the games became more enjoyable over time (average 3.67 to 4.36), with no significant differences ($p = 0.4352$). About half of the players of *WaveRider* stated that they were willing to continue playing the game actively (average 3.15).

4.3. RQ3: How game concept and genre affect exercise and pulse?

Table 6 shows responses related to exercise. Most players felt they were exercising while playing (average 4.08 +). The *FlipBoard* survey included a statement about thinking of being tired, averaging 3.67. The intensity levels of *Lane Rider* and *2D Boss Fighter* were above medium (average 3.58) and high (average 4.47), respectively. Specific statements for each game indicated positive exercise feedback, with *Lane Rider* and *2D Boss Fighter* players feeling they got a good workout. Further, that *2D Boss Fighter* players responded with an average 4.25 to the statement "Playing this game is more fun than just exercising on an exercise bike".

The experiments aimed to assess whether exergames could provide beneficial physical exercise. For significant health benefits, adults should engage in at least 150 min of moderate-intensity or 75 min of vigorous-intensity aerobic physical activity weekly [67]. Moderate intensity requires a heart rate of 50–70 % of maximum, while vigorous

Table 6

Feedback Related to Exercise Scale: 1: Fully Disagree, 2: Disagree, 3: Neutral, 4: Agree, 5: Fully Agree.

Topic	Lane Rider		2D Boss Fighter		WaveRider		FlipBoard	
	Avr	Std	Avr	Std	Avr	Std	Avr	Std
Exercise: I felt I was exercising	4.33	0.65	4.33	0.69	4.08	0.75	NA	NA
Exercise: The game had a high intensity level	4.47	0.50	3.58	0.67	NA	NA	NA	NA

intensity requires 70–80 %. High-Intensity Interval Training (HIT) should reach 80 % or more to maximize health outcome [68]. Maximum heart rate can be estimated as [69]:

$$HR_{Max} = 211 - 0.64 * age.$$

For an average age of 24 as was the case for these studies, HR_{Max} is approximately 193 bpm. Thus, *moderate exercise* requires a pulse of at least 96.6 bpm, *vigorous exercise* requires 134.6 bpm, and *HIT requires* 154.6 bpm. Table 7 presents the measured heart rates from the second game section across the four games.

The table indicates the range of heart rates across different intensity profiles (lowest, median, highest) for each game. It shows the average heart rate throughout the game session, the maximum heart rate, and the heart rate during the most intense part of the session (80 %+ heart

Table 7

Measured Heartrates from the Second Game Sessions of the Four Games.

Player profile	Game	HR _{Avr}	HR _{Max}	HR _{80% Perc}
Lowest exercise intensity	Lane Rider	108.9	130.0	117.0
	2D Boss Fighter	107.5	126.0	114.0
	WaveRider	123.4	148.0	135.0
	FlipBoard	107.5	126.0	114.0
Median exercise intensity	Lane Rider	133.0	179.0	161.1
	2D Boss Fighter	132.4	175.0	161.0
	WaveRider	141.3	173.0	163.0
	FlipBoard	133.7	153.0	140.0
Highest exercise intensity	Lane Rider	165.2	186.0	174.5
	2D Boss Fighter	176.9	200.0	194.0
	WaveRider	162.9	206.0	194.0
	FlipBoard	154.3	178.0	162.0

Table 5

Results Related to Enjoyment, Motivation and Replayability Scale: 1: Fully Disagree, 2: Disagree, 3: Neutral, 4: Agree, 5: Fully Agree.

Topic	Lane Rider		2D Boss Fighter		WaveRider		FlipBoard	
	Avr	Std	Avr	Std	Avr	Std	Avr	Std
Enjoyment: I had fun while playing the game	4.33	0.51	4.08	0.40	4.15	0.78	4.44	0.53
Motivation: Motivation to exercise increased from playing the game	3.53	1.15	3.75	0.54	3.77	1.21	NA	NA
Replayability: I would like to play the game again	4.60	0.50	3.83	0.77	3.77	1.08	4.22	0.46
Replayability: Game got more enjoyable over time	4.36	0.66	3.67	1.08	4.31	0.52	3.67	1.31

rate). Overall, the heart rates suggest that the games varied in intensity but generally provided a good workout. However, there was a significant difference in heart rate among players with the lowest, median, and highest exercise intensities. The heart rate of players was influenced by their fitness levels, pedaling speed, and chosen resistance on the exercise bike. Participants were allowed to freely choose their resistance levels. The results show no significant difference in heart rates between the four games.

Table 8 presents an overview of the percentage of players from the four games who reached the heart rates required for vigorous (70 %+ of HR_{Max}) and HIT (80 %+ of HR_{Max}) exercise intensity. All players experienced average heart rates above the threshold for moderate exercise intensity (50 %+ of HR_{Max}), with the lowest heart rate measured at 107.5 bpm. Furthermore, 30 % to 62 % of subjects reached an average heart rate that qualifies for vigorous exercise intensity. The *FlipBoard* game had the fewest players reaching vigorous intensity, while *WaveRider* had the most. For the most intense part of the sessions (with 80 % highest heart rates), 60 % of *Lane Rider* and *FlipBoard* players reached vigorous intensity, and 100 % of *2D Boss Fighter* and *WaveRider* players did so. Accordingly, 8–15 % of subjects achieved HIT intensity for average heart rate, and 10–85 % for the 80 % or more intense part of the workout. Based on the results, *2D Boss Fighter* and *WaveRider* were more effective for HIT compared to *Lane Rider* and *FlipBoard*.

5. Discussion

This section includes a discussion of the results as well as addressing the threats to validity of the study.

5.1. Discussion of the results

The experiments demonstrated that varying game concepts and genres significantly impacted gameflow, enjoyment, motivation, replayability, exercise, and heart rate intensity. Key findings include the following:

- All games positively influenced the eight gameflow elements, with *WaveRider* showing higher **immersion** due to its music-linked gameplay. This positive effect of music-linked gameplay is also found in games like *Dance Dance Revolution* [15], *Beat Saber* [29], and *Peloton Lanebreak*.
- Players of *WaveRider* and *FlipBoard* felt a greater sense of **skill** development compared to *Lane Rider*, likely due to the more conventional controls and gameplay in *Lane Rider* that did not require extensive skill development. Interestingly, despite the variation in game controls, there was no significant difference in players' feedback on **control**. However, players of *2D Boss Fighter* felt less in control, likely due to its unique and complex controls.
- In the related work section, exerbike games demonstrated a range of control schemes. For instance, in *Zwift* [12], *Peloton Lanebreak* [40], and *Play'n Ride* [33], player input was limited to pedaling speed alone. In contrast, *PaperDude* [32] and the study by Shaw et al. [35] integrated pedaling sensors with VR and camera tracking, while Monedero et al. [37] and *Life is a Village* [38] combined pedaling

sensors with PS2 controllers. The **control scheme** in the games presented in this study is similar in that it combines a pedaling sensor with additional input devices. However, it is distinct in its use of custom-designed buttons mounted directly on the handlebars of the exercise bike. This design integrates gameplay controls more seamlessly into the exercise equipment, potentially enhancing the user experience by making interactions feel like a natural extension of the physical activity (see Fig. 1 and Fig. 3).

- Varying the **function of pedaling** within gameplay allowed our study to explore different game genres and examine how these variations influenced player experience. In our games, pedaling controlled forward movement in *Lane Rider*, firing and rotor speed in *2D Boss Fighter*, the vertical position of a tone indicator in *WaveRider*, and the rotation angle of a platform in *FlipBoard*. Importantly, our findings indicate that diversifying the role of pedaling did not lead to any significant adverse effects on gameplay or player engagement. This contrasts with most related exerbike games, where pedaling typically translates to forward movement, with a few notable exceptions – such as *Play'n Ride* [33], where pedaling controlled the altitude of a flying character, and *Greedy Rabbit* [31], which used pedaling to adjust movement speed in a 2D environment.
- **Social interaction** was an important source of gameflow. Games that offered different modes of interaction showed that players preferred competitive play (PvP) over collaborative play (PvE) and multiplayer over single-player. Similar preferences were observed in games like *Pokémon Go* [6,22] and *Zwift* [39].
- The **game concept** and genre did not significantly affect enjoyment, motivation, and replayability. All games exhibited high levels of **enjoyment**, a moderate increase in players' **motivation** to exercise, and mostly positive effects on **replayability**, with *Lane Rider* and *FlipBoard* slightly more favored in terms of replayability. Other studies have also found exergames to positively impact exercise motivation [6,16,22,27,28], but this effect only lasts as long as players continue playing [70]. This means that replayability and a game that can last for a long time is key to success.
- In terms of **exercise intensity**, all four games provided effective workouts, with many players reaching vigorous and high-intensity interval training (HIT) levels. All participants experienced average heart rates above *moderate intensity*, indicating that playing these games for at least 150 min per week would meet the recommended physical activity levels for significant health benefits [67]. For *2D Boss Fighter* and *WaveRider*, more than half of the players had average heart rates above *vigorous intensity*, suggesting that 75 min of gameplay per week would be sufficient for significant health benefits [67].
- Previous experiments have shown that among non-bike-based games, only *Dance Dance Revolution* can reach moderate intensity at standard difficulty and vigorous intensity at hard difficulty [71,72]. *Ring Fit Adventure* is another non-bike exergame where players can achieve moderate intensity, with the added benefit of increased physical strength [73]. However, for HIT exercises or similar high-intensity workouts, only bike-based exergames are effective. The exercise intensity in these games depends on the player's willingness to push their physical limits in terms of pedaling speed and resistance.

These insights suggest that well-designed exergames can provide enjoyable and effective ways to achieve recommended physical activity levels by leveraging game design elements to enhance player engagement and exercise intensity. However, the primary challenge is sustaining player motivation over time. For instance, studies of *Pokémon Go* demonstrated that players maintained higher physical activity levels for several months while playing, but lost all health benefits after they stopped [74].

Table 8
Percentage of Subjects Reaching Vigorous and HIT Exercise Intensity.

Game	Vigorous (135 bpm +)			HIT (154 bpm +)		
	HR_{Avr}	$HR_{80\%}$ Perc	HR_{Max}	HR_{Avr}	$HR_{80\%}$ Perc	HR_{Max}
Lane Rider	40 %	60 %	80 %	13 %	47 %	60 %
2D Boss Fighter	50 %	100 %	100 %	8 %	75 %	92 %
WaveRider	62 %	100 %	100 %	15 %	85 %	92 %
FlipBoard	30 %	60 %	80 %	10 %	10 %	40 %

5.2. Recommendations for pedal exercise games

Based on the results and discussion from our *meta*-experiment involving four pedal-based exercise games, we propose the following design recommendations:

1. **Design a Well-Balanced Game with Strong Gameflow and Dual-Flow Integration:** Creating an effective exergame requires a careful balance between engaging gameplay and appropriate physical challenge. A well-designed gameflow enhances immersion, which can help shift the player's attention away from physical exertion, thereby improving the overall exercise experience [75]. Our study demonstrated positive evaluations across all eight gameflow elements, as well as favorable feedback on the exercise component of the games. To support a wide range of players, both game mechanics and physical demands should be adaptable – enabling players to progress according to their individual skill and fitness levels. While the games in our study offered limited options for adjusting game difficulty, they did support variation in physical challenge by allowing players to modify pedaling speed and resistance. This led to a range of exercise intensities, from moderate to vigorous, accommodating different physical capabilities and preferences.
2. **Diversify and Effectively Teach Game Controls:** In exergames utilizing an exercise bike, controls must be both intuitive for beginners and allow for skill development over time. Our findings demonstrate that pedaling input can serve multiple gameplay functions. However, it is essential that these controls are introduced in an engaging and accessible way. All four games in our study were rated as highly enjoyable and generally achieved strong gameflow. Nevertheless, games with more complex control schemes showed reduced gameflow, particularly in elements related to *player skill* and *control*. This highlights the importance of balancing control complexity with effective onboarding and learning mechanisms.
3. **Design Games to Explicitly Motivate Exercise:** Our study showed that all games encouraged players to play and replay them. This indicates that the games implicitly motivated physical activity, as each provided a solid workout. However, we also identified challenges in designing games that explicitly promote sustained exercise. One potential improvement is to visualize the player's physical progress and health impact. This feature was not included in our games but is used effectively in *Zwift* [12,39] and *Peloton Lanebreak* [40]. Progress can be shown using statistics, graphics, animated plots, or images [51]. These elements help players see the connection between gameplay and their physical improvement.
4. **Ensure Long-Term Engagement:** Research shows that the health benefits of exergames last only as long as players remain engaged [6,74]. For example, studies on *Pokémon Go* found a small increase in physical activity among youth and young adults for less than two months, and up to seven months for older adults [76]. This highlights the need for exergames to offer high replayability and long-term engagement. Players who continued playing *Pokémon Go* often cited fun, immersive gameplay [70,77,78], and social interaction [79,80] as key motivators. In our study, the use of prototypes limited replayability. To support sustained engagement, designers should consider features such as user-generated content, evergreen gameplay elements, and procedurally generated levels and challenges [81].
5. **Integrate Diverse Reward Systems:** Well-designed rewards can significantly boost player retention and motivation. In our study, the games featured limited reward systems, mainly focusing on scores, achievements, and basic feedback messages. In contrast, *Ring Fit Adventure* provides a strong example of effective reward design. It includes scores, experience points (XP), avatar upgrades, item and resource rewards, achievements, feedback messages, plot-related visuals, and unlockable content [51]. These elements work together to keep players engaged and motivated to progress.

Pokémon Go, arguably the most impactful exergame in terms of global health, also uses a wide range of reward systems. Among the most motivating are XP-based progression, leveling up, avatar customization, and unlockable features [70,82,83].

6. **Encourage High-Intensity Exercise:** In our study, exercise intensity varied across players and games. The aim is to design gameplay that keeps players engaged and unaware of how intensely they are pedaling, while also incorporating periods of lower intensity. One effective strategy is to simulate high-intensity interval training (HIT) [84], with alternating bursts of intense effort and recovery phases. *Pedal Tanks* is a strong example of this approach [44]. It motivates players to perform HIT naturally, without explicit instruction [45]. For example, players pedal at a moderate pace while searching for enemy tanks. However, during flag captures or escapes, they instinctively pedal at high intensity due to the game's pressure and excitement.
7. **Include audio and social features:** While all games in our study featured audio, only *WaveRider* explicitly integrated music with player performance. As players improved, new musical layers were unlocked, resulting in a progressively richer auditory experience – an approach that led to a statistically significant increase in immersion compared to the other games. Social interaction also played a key role, with 69 % of *WaveRider* players preferring the competitive mode. This preference aligns with findings from a study on *Kahoot!* [66], which demonstrated that competitive play significantly enhanced concentration, engagement, enjoyment, and motivation. Furthermore, the *Kahoot!* study highlighted music's contribution to a socially engaging and lively atmosphere. Similarly, prior research on pedal-based exergames [38] supports the idea that gameplay-integrated music and social interaction substantially increase player enjoyment.

5.3. Threats to validity

This study is a *meta*-analysis of four quasi-experiments, each with potential threats to validity. A key limitation is the absence of a control group and the lack of direct comparisons to alternative approaches or games. Additionally, the sample size was relatively small, with a total of 53 participants across the four experiments (ranging from 10 to 16 participants per experiment). A larger sample would have allowed for more robust statistical analysis, including subgroup comparisons based on gender, age, gaming experience, and physical activity levels. While the participants (aged 21–50) were representative of typical players, a broader age range would improve the generalizability of the findings.

To mitigate the impact of the small sample size, the study triangulated data from questionnaires, observations, and interviews, enhancing the reliability of the findings. Additionally, the results were compared with related research to further validate key insights.

Another limitation is that most participants were physically active and played video games regularly, which may influence their engagement with exergames. Future research should explore how baseline physical activity and gaming habits affect exergame usage and player experiences.

A further threat to validity is the non-standardized nature of the questionnaires used in the four experiments. The survey design was based on the gameflow framework and Malone's theory of intrinsic motivation, but variations in the statements across experiments were noted and addressed when presenting the results.

Finally, differences in heart rate measurement tools pose a potential validity concern. While Polar H10 heart rate monitors were used in three experiments, the first experiment (*Lane Rider*) relied on the Fitbit Charge 3, which is considered less accurate. The use of different devices was due to the unavailability of the Polar H10 at the time of the first experiment.

6. Conclusion

This *meta-study* examined and compared how four different pedal-based exergames influenced gameflow, enjoyment, motivation, replayability, pulse, and exercise experience. While the games varied in genre, gameplay, and controls, all were played on the same PlayPulse exercise bike platform.

The findings showed that all four games positively influenced the eight gameflow elements (RQ1). Notably, *WaveRider* demonstrated higher immersion due to its music-linked gameplay. Unique game controls enhanced the sense of skill development, and the integration of pedaling mechanics was found to be critical for engagement. Additionally, social interaction played a key role in enhancing gameflow in pedal-based exergames.

Regarding enjoyment, motivation, and replayability (RQ2), the results indicated no significant differences across genres. All games produced high enjoyment levels, moderate increases in motivation to exercise, and mostly positive effects on replayability. However, *Lane Rider* and *FlipBoard* showed a tendency toward higher replayability scores.

In terms of exercise effectiveness (RQ3), all four games enabled players to reach vigorous and high-intensity interval training levels, suggesting that playing at least three times per week could provide significant health benefits. Many participants reported being aware of the physical exertion while playing and noted the intensity of the workouts.

Finally, this study proposed seven recommendations for designing pedal-based exergames that maximize player enjoyment while ensuring effective physical exercise. Future research should further explore and validate the long-term impact of these recommendations.

Declaration of generative AI and AI-assisted technologies in the writing process

The only use of generative AI in this paper was for spelling and grammar improvements using ChatGPT.

CRediT authorship contribution statement

Alf Inge Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Emil Petter Schröder:** Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Øystein Hammersland:** Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Christian Magnus Rand:** Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sander Kaalaas Østvik:** Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Aschmirthan Sivarajan:** Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sondre Strand Haltbakk:** Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sigrid Greiff Elnan:** Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Vilde Voss Stabell:** Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] A. López-Valenciano, X. Mayo, G. Liguori, R. Copeland, M. Lamb, A. Jimenez, Changes in sedentary behaviour in European union adults between 2002 and 2017, *BMC Public Health* 20 (2020) 1–10, <https://doi.org/10.1186/s12889-020-09293-1>.
- [2] I.-M. Lee, E.J. Shiroma, F. Lobelo, P. Puska, S.N. Blair, P.T. Katzmarzyk, Effect of physical inactivity on major non-communicable diseases worldwide: an analysis of burden of disease and life expectancy, *Lancet* 380 (2012) 219–229, [https://doi.org/10.1016/S0140-6736\(12\)61031-9](https://doi.org/10.1016/S0140-6736(12)61031-9).
- [3] A. Biswas, P.I. Oh, G.E. Faulkner, R.R. Bajaj, M.A. Silver, M.S. Mitchell, D.A. Alter, Sedentary time and its association with risk for disease incidence, mortality, and hospitalization in adults: a systematic review and meta-analysis, *Ann. Intern. Med.* 162 (2015) 123–132, <https://doi.org/10.7326/M14-1651>.
- [4] B.K. Clark, T. Sugiyama, G.N. Healy, J. Salmon, D.W. Dunstan, N. Owen, Validity and reliability of measures of television viewing time and other non-occupational sedentary behaviour of adults: a review, *Obes. Rev.* 10 (2009) 7–16, <https://doi.org/10.1111/j.1467-789X.2008.00508.x>.
- [5] R.M. Ryan, E.L. Deci, Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being, *Am. Psychol.* 55 (2000) 68, <https://doi.org/10.1037/0003-066X.55.1.68>.
- [6] A.I. Wang, Systematic literature review on health effects of playing pokémon go, *Entertain. Comput.* (2021) 100411, <https://doi.org/10.1016/j.entcom.2021.100411>.
- [7] Y. Xian, H. Xu, H. Xu, L. Liang, A.F. Hernandez, T.Y. Wang, E.D. Peterson, An initial evaluation of the impact of Pokémon GO on physical activity, *J. Am. Heart Assoc.* 6 (2017) e005341, <https://doi.org/10.1161/JAHA.116.005341>.
- [8] W. Liu, A. Ligmman-Zielinska, A pilot study of Pokémon Go and players' physical activity, *Games for Health Journal* 6 (2017) 343–350, <https://doi.org/10.1089/g4h.2017.0036>.
- [9] B. Jungselius, A. Weilenmann, M. Rost, Pokémon GO and Mobile Wellbeing: initial Observations on Experiences and Reported connection, *Data Science for Health*, 2015.
- [10] J.E. Barkley, A. Lepp, E.L. Glickman, Pokémon go! may promote walking, discourage sedentary behavior in college students, *Games for Health Journal* 6 (2017) 165–170, <https://doi.org/10.1089/g4h.2017.0009>.
- [11] D.J. Mateo, Pokémon GO may increase physical activity and decrease sedentary behaviors regular physical, *Am. J. Public Health* 107 (2017) 37, <https://doi.org/10.2105/AJPH.2016.303532>.
- [12] D. Westmattmann, J.-G. Grotenhermen, B. Stoffers, G. Schewe, Exploring the adoption of mixed-reality sport platforms: a qualitative study on ZWIFT, *European Conference on Information Systems (ECIS)*, Marrakech, Marokko (2021).
- [13] I. Bogost, *Persuasive games: the expressive power of videogames*, MIT Press, 2010.
- [14] Y. Oh, S. Yang, Defining exergames & exergaming, *Proceedings of Meaningful Play* (2010) 1–17.
- [15] J. Hoysniemi, International survey on the Dance Dance Revolution game, *Computers in Entertainment (CIE)*, 4 (2006) 8-es. 10.1145/1129006.1129019.
- [16] A.D. Kloos, N.E. Fritz, S.K. Kostyk, G.S. Young, D.A. Kegelmeyer, Video game play (Dance Dance Revolution) as a potential exercise therapy in Huntington's disease: a controlled clinical trial, *Clin. Rehabil.* 27 (2013) 972–982, <https://doi.org/10.1177/0269215513487235>.
- [17] L.-Y. Chuang, H.-Y. Hung, C.-J. Huang, Y.-K. Chang, T.-M. Hung, A 3-month intervention of Dance Dance Revolution improves interference control in elderly females: a preliminary investigation, *Exp. Brain Res.* 233 (2015) 1181–1188, <https://doi.org/10.1007/s00221-015-4196-x>.
- [18] H. Liu, Y. Xing, Y. Wu, Effect of wii fit exercise with balance and lower limb muscle strength in older adults: a meta-analysis, *Front. Med.* 9 (2022) 812570, <https://doi.org/10.3389/fmed.2022.812570>.
- [19] B.L. Cone, S.S. Levy, D.J. Goble, Wii Fit exergame training improves sensory weighting and dynamic balance in healthy young adults, *Gait Posture* 41 (2015) 711–715, <https://doi.org/10.1016/j.gaitpost.2015.01.030>.
- [20] L.E. Graves, N.D. Ridgers, K. Williams, G. Stratton, G. Atkinson, N.T. Cable, The physiological cost and enjoyment of Wii Fit in adolescents, young adults, and older adults, *J. Phys. Act. Health* 7 (2010) 393–401, <https://doi.org/10.1123/jpah.7.3.393>.
- [21] J.R. Franco, K. Jacobs, C. Inzerillo, J. Kluzik, The effect of the Nintendo Wii Fit and exercise in improving balance and quality of life in community dwelling elders, *Technol. Health Care* 20 (2012) 95–115, <https://doi.org/10.3233/thc-2011-0661>.
- [22] A.I. Wang, A. Skjervold, Health and social impacts of playing pokémon go on various player groups, *Entertain. Comput.* 39 (2021) 100443, <https://doi.org/10.1016/j.entcom.2021.100443>.
- [23] A.I. Wang, K. Hagen, T. Høivik, G.M. Olsen, Evaluation of the Game Exermon - a Strength Exergame Inspired by Pokémon Go, in: 14th International Conference on Advances in Computer Entertainment Technology (ACE 2017), Springer, London, UK, 2017. 10.1007/978-3-319-76270-8_27.
- [24] Y.-S. Wu, W.-Y. Wang, T.-C. Chan, Y.-L. Chiu, H.-C. Lin, Y.-T. Chang, H.-Y. Wu, T.-C. Liu, Y.-C. Chuang, J. Wu, Effect of the Nintendo Ring Fit Adventure exergame on running completion time and psychological factors among university students engaging in distance learning during the COVID-19 pandemic: randomized

- controlled trial, *JMIR Serious Games* 10 (2022) e35040, <https://doi.org/10.2196/35040>.
- [25] K. Suire, K. Spring, C. Jones, Get fit with ring-fit adventure™: physiological outcomes of a novel active video game intervention among women, *J. Womens Health Care Manage* 3 (2022) 1–5, <https://doi.org/10.47275/2692-0948-135>.
- [26] T. Sato, K. Shimizu, Y. Shiko, Y. Kawasaki, S. Orita, K. Inage, Y. Shiga, M. Suzuki, M. Sato, K. Enomoto, Effects of nintendo ring fit adventure exergame on pain and psychological factors in patients with chronic low back pain, *Games for Health Journal* 10 (2021) 158–164, <https://doi.org/10.1089/g4h.2020.0180>.
- [27] T. Thai, The Influence of Exergaming on Heart Rate, Perceived Exertion, Motivation to Exercise, And Time Spent Exercising, in: *Sport and Movement Science*, Salem State University, 2019.
- [28] O. Kivelä, P. Alaves, A. Visuri, T. Ojala, Study on the Motivational and Physical Effects of two VR Exergames, in: 2019 11th International Conference on Virtual Worlds and Games for Serious Applications (VS-Games), IEEE, Vienna, Austria, 2019, pp. 1–2, 10.1109/VS-Games.2019.8864544.
- [29] J.-H. Tammy Lin, D.-Y. Wu, N. Bowman, Beat saber as virtual reality exercising in 360 degrees: a moderated mediation model of VR playable angles on physiological and psychological outcomes, *Media Psychol.* 26 (2023) 414–435, <https://doi.org/10.1080/15213269.2022.2154806>.
- [30] M. Pau, M. Porta, R. Berton, F.G.M. Mattos, E. Cocco, D. Cattaneo, Effect of immersive virtual reality training on hand-to-mouth task performance in people with multiple sclerosis: a quantitative kinematic study, *Mult. Scler. Relat. Disord.* 69 (2023) 104455, <https://doi.org/10.1016/j.msard.2022.104455>.
- [31] D. Pasco, C. Roure, G. Kermarrec, Z. Pope, Z. Gao, The effects of a bike active video game on players' physical activity and motivation, *J. Sport Health Sci.* 6 (2017) 25–32, <https://doi.org/10.1016/j.jshs.2016.11.007>.
- [32] J. Bolton, M. Lambert, D. Lirette, B. Unsworth, PaperDude: A Virtual Reality Cycling Exergame, in: CHI'14 Extended Abstracts on Human Factors in Computing Systems, 2014, pp. 475–478.
- [33] C.P. Lelardeux, M. Galaup, F. Hall, P. Lagarrigue, Designing a Fantasy Bike-based Exergame to Foster Physical Activity, *International Journal of Serious Games*, 7 (2020) 21–39, 10.17083/ijsg.v7i3.362.
- [34] R. Al-Hrathi, A. Karime, H. Al-Osman, A. El Saddik, Exerlearn Bike: an Exergaming System for Children's Educational and Physical Well-being, in: 2012 IEEE International Conference on Multimedia and Expo Workshops, 2012, pp. 489–494, <https://doi.org/10.1109/ICMEW.2012.91>.
- [35] L.A. Shaw, J. Buckley, P.M. Corballis, C. Lutteroth, B.C. Wünsche, Competition and cooperation with virtual players in an exergame, *PeerJ Comput. Sci.* 2 (2016) e92.
- [36] L.A. Shaw, B.C. Wünsche, C. Lutteroth, S. Marks, J. Buckley, P. Corballis, Development and Evaluation of an Exercycle Game using Immersive Technologies, in: A. Maeder, J. Warren (Eds.), *Research and Practice in Information Technology (CRPIT)*, Australian Computer Society, Sydney, Australia, 2015, pp. 75–85, <https://hdl.handle.net/10292/8786>.
- [37] J. Monedero, E.J. Lyons, D.J. O'Gorman, Interactive video game cycling leads to higher energy expenditure and is more enjoyable than conventional exercise in adults, *PLoS One* 10 (2015) e0118470, <https://doi.org/10.1371/journal.pone.0118470>.
- [38] J. Yim, T.N. Graham, Using Games to increase Exercise Motivation, in: *Proceedings of the 2007 Conference on Future Play*, 2007, pp. 166–173, <https://doi.org/10.1145/1328202.1328232>.
- [39] A. Devine, F. Devine, A. Burns, An examination of the virtual event experience of cyclists competing on zwift, *Event Management* 28 (2024) 151–167, <https://doi.org/10.3727/152599523x16907613842110>.
- [40] Peloton, Peleton Lanebreak, A Gamified Workout Experience, in, Web: <https://www.onepeloton.com/lanebreak>, 2025.
- [41] K. Hagen, S. Weie, Creating an Effective Exergame with Lasting Entertainment Value, in, *Computer Science*, Norwegian University of Science and Technology, NTNU Trondheim, 2015.
- [42] K. Hagen, S. Weie, K. Chorianopoulos, A.I. Wang, L. Jaccheri, Pedal Tanks, in: *International Conference on Entertainment Computing*, Springer, 2015, pp. 539–544, 10.1007/978-3-319-24589-8_53.
- [43] M. Mora-Cantallos, M.-Á. Sicilia, MOBA games: a literature review, *Entertain. Comput.* 26 (2018) 128–138, <https://doi.org/10.1016/j.entcom.2018.02.005>.
- [44] K. Hagen, K. Chorianopoulos, A.I. Wang, L. Jaccheri, S. Weie, Gameplay as Exercise, in: *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*, 2016, pp. 1872–1878, <https://doi.org/10.1145/2851581.2892515>.
- [45] T. Moholdt, S. Weie, K. Chorianopoulos, A.I. Wang, K. Hagen, Exergaming can be an innovative way of enjoyable high-intensity interval training, *BMJ Open Sport Exerc. Med.* 3 (2017) e000258, <https://doi.org/10.1136/bmjsem-2017-000258>.
- [46] J. Berg, G. Haugen, A.I. Wang, T. Moholdt, High-intensity exergaming for improved cardiorespiratory fitness: a randomised, controlled trial, *Eur. J. Sport Sci.* (2021) 1–10, <https://doi.org/10.1080/17461391.2021.1921852>.
- [47] A. Soria Campo, A.I. Wang, T. Moholdt, J. Berg, Physiological and perceptual responses to single-player vs. multiplayer exergaming, *Front. Sports Active Living* 4 (2022) 903300, <https://doi.org/10.3389/fspor.2022.903300>.
- [48] J. Berg, T. Moholdt, Game on: a cycling exergame can elicit moderate-to-vigorous intensity, A Pilot Study, *BMJ Open Sport & Exercise Medicine* 6 (2020) e000744, <https://hdl.handle.net/11250/2734468>.
- [49] T.W. Malone, Toward a theory of intrinsically motivating instruction*, *Cognit. Sci.* 5 (1981) 333–369, <https://doi.org/10.1207/s15516709cog0504.2>.
- [50] P. Sweetser, P. Wyeth, GameFlow: a model for evaluating player enjoyment in games, *ACM Comput. Entertain.* 3 (2005), <https://doi.org/10.1145/1077246.1077253>.
- [51] H. Wang, C.-T. Sun, Game Reward Systems: Gaming Experiences and Social Meanings, in: *DiGRA conference*, DiGRA, Tampere, Finland, 2011. 10.26503/dl.v2011i1.594.
- [52] J. Sinclair, P. Hingston, M. Masek, Exergame Development using the dual Flow Model, in: *Proceedings of the Sixth Australasian Conference on Interactive Entertainment*, 2009, pp. 1–7, <https://doi.org/10.1145/1746050.1746061>.
- [53] Wikipedia, Temple Run, in, Web: https://en.wikipedia.org/wiki/Temple_Run, 2025.
- [54] Wikipedia, Subway Surfers, in, Web: https://en.wikipedia.org/wiki/Subway_Surfers, 2025.
- [55] Ø. Hammersland, E.P. Schröder, Lane Rider-an Exploratory Study on the Benefits of Exergaming, Master Thesis, Norwegian University of Science and Technology, NTNU Open, 2020.
- [56] C.M. Rand, S.K. Østvik, 2D Boss Fighter Exergame-The Creation and Evaluation of an Exercise Bike Game to increase Motivation for Physical activity, Master Thesis, Norwegian University of Science and Technology, NTNU Open, 2021.
- [57] A. Wood, A. Summerville, Understanding Boss Battles: A Case Study of Cuphead, Exag unbekannt: <https://exag.org/archive/wood2019battles.pdf>, (2019).
- [58] Wikipedia, SingStar, in, Web: <https://en.wikipedia.org/wiki/SingStar>, 2025.
- [59] Wikipedia, Dance Dance Revolution, in, Web: https://en.wikipedia.org/wiki/Dance_Dance_Revolution, 2025.
- [60] Wikipedia, Guitar Hero, in, Web: https://en.wikipedia.org/wiki/Guitar_Hero, 2025.
- [61] A. Sivaranjan, S.S. Haltbakk, WaveRider-A High-intensity Interval Training Exergame for Exercise Bikes, Master Thesis, Norwegian University of Science and Technology, NTNU Open, 2022.
- [62] S.G. Elnan, V.V. Stabell, FlipBoard-Unexplored Genres in Exergames, in: Master Thesis, Norwegian University of Science and Technology, NTNU Open, 2023. <https://hdl.handle.net/11250/3095623>.
- [63] V.R. Basili, Software Modeling and Measurement: the Goal/Question/Metric Paradigm, University of Maryland for Advanced Computer Studies, 1992.
- [64] H.B. Mann, D.R. Whitney, On a test of whether one of two random variables is stochastically larger than the other, *Ann. Math. Stat.* 18 (1947) 50–60.
- [65] W.H. Kruskal, W.A. Wallis, Use of ranks in one-criterion variance analysis, *J. Am. Stat. Assoc.* 47 (1952) 583–621, <https://doi.org/10.2307/2280779>.
- [66] A.I. Wang, A. Lieberoth, The Effect of Points and Audio on Concentration, Engagement, Enjoyment, Learning, Motivation, and Classroom Dynamics using Kahoot!, in: 10th European Conference of Games Based Learning, Academic Conferences and Publishing International Limited, Paisley, Scotland, 2016.
- [67] R.D. Olson, K.L. Piercy, R.P. Troiano, R.M. Ballard, J.E. Fulton, D.A. Galauska, S. Pfohl, Y., A. Vaux-Bjerke, J.B. Quam, S.M. George, K. Sprow, S.A. Carlson, E.T. Hyde, K. Olscamp, Physical Activity Guidelines for Americans, in, Department of Health & Human Services, USA, Web: https://health.gov/sites/default/files/2019-09/Physical_Activity_Guidelines_2nd_edition.pdf, 2018.
- [68] T. Karlsen, I.-L. Aamot, M. Haykowsky, Ø. Rognmo, High intensity interval training for maximizing health outcomes, *Prog. Cardiovasc. Dis.* 60 (2017) 67–77, <https://doi.org/10.1016/j.pcad.2017.03.006>.
- [69] B.M. Nes, I. Janszky, U. Wisløff, A. Støylen, T. Karlsen, Age-predicted maximal heart rate in healthy subjects: the HUNT fitness study, *Scand. J. Med. Sci. Sports* 23 (2013) 697–704, <https://doi.org/10.1111/j.1600-0838.2012.01445.x>.
- [70] P. Rasche, A. Schlomann, A. Mertens, Who is still playing Pokémon Go? a web-based survey, *JMIR Serious Games* 5 (2017), <https://doi.org/10.2196/games.7197>.
- [71] A. Norlin, J.P. Porcari, B. Udermann, K. Mally, C. Dodge, C. Foster, Exercise intensity of dance dance revolution: exercise physiology; exercise training: poster# 75, *J. Cardiopulm. Rehabil. Prev.* 27 (2007) 331, <https://doi.org/10.1097/01.HCR.0000291341.61795.01>.
- [72] A.E. Maloney, K.A. Threlkeld, W.L. Cook, Comparative effectiveness of a 12-week physical activity intervention for overweight and obese youth: exergaming with “dance dance revolution”, *GAMES FOR HEALTH: Research, Development, and Clinical Applications* 1 (2012) 96–103, <https://doi.org/10.1089/g4h.2011.0009>.
- [73] P.B. Moreno Carbajal, N.E. Corrales Camargo, R. Espinoza-Gutiérrez, J.J. Calleja-Núñez, B. Montero-Herrera, J. Cerna, J.A. Aburto-Corona, Energy expenditure validation of an exergame platform: ring fit adventure use in adults with overweight and obesity, *Games for Health Journal* 13 (2024) 288–296, <https://doi.org/10.1089/g4h.2023.0159>.
- [74] A. Gabbiadini, C. Sagioglou, T. Greitemeyer, Does Pokémon Go lead to a more physically active life style? *Comput. Hum. Behav.* 84 (2018) 258–263, <https://doi.org/10.1016/j.chb.2018.03.005>.
- [75] J. Sinclair, P. Hingston, M. Masek, Considerations for the Design of Exergames, in: 5th International Conference on Computer Graphics and Interactive Techniques in Australia and Southeast Asia, 2007, pp. 289–295, <https://doi.org/10.1145/1321261.1321313>.
- [76] T. Baranowski, E.J. Lyons, Scoping review of pokemon go: comprehensive assessment of augmented reality for physical activity change, *Games for Health Journal* 9 (2020) 71–84, <https://doi.org/10.1089/g4h.2019.0034>.
- [77] F.J. Zach, I.P. Tussyadiah, To Catch Them All—The (Un) intended Consequences of Pokémon GO on Mobility, Consumption, and Wellbeing, in: *Information and Communication Technologies in Tourism 2017*, 2017, pp. 217–227.
- [78] A. Meschtscherjakov, S. Trösterer, A. Lupp, M. Tscheligi, W.A.L.K. Pokémon, Persuasive Effects of Pokémon GO Game-design elements, in: *International Conference on Persuasive Technology*, 2017, pp. 241–252, https://doi.org/10.1007/978-3-319-55134-0_19.
- [79] L.K. Militello, N. Hanna, C.R. Nigg, Pokémon GO within the context of family health: retrospective study, *J. Med. Internet Res.* 20 (2018), <https://doi.org/10.2196/10679>.

- [80] J.A. Bonus, A. Peebles, M.-L. Mares, I.G. Sarmiento, Look on the bright side (of Media Effects): Pokémon Go as a catalyst for positive life experiences, *Media Psychol.* 21 (2018) 263–287, <https://doi.org/10.1080/15213269.2017.1305280>.
- [81] M. Hendrikx, S. Meijer, J. Van Der Velden, A. Iosup, Procedural Content Generation for Games: A Survey, *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)*, 9 (2013) 1–22. 10.1145/2422956.2422957.
- [82] L. Evans, M. Saker, The Playeur and Pokémon Go: examining the effects of locative play on spatiality and sociability, *Mobile Media and Communication* (2018), <https://doi.org/10.1177/2050157918798866>.
- [83] P. Loveday, J. Burgess, Flow and Pokémon GO: the contribution of game level playing alone, and nostalgia to the flow state, *E-Journal of Social & Behavioural Research in Business* 8 (2017) 16–28.
- [84] P.B. Laursen, D.G. Jenkins, The scientific basis for high-intensity interval training, *Sports Med.* 32 (2002) 53–73, <https://doi.org/10.2165/00007256-200232010-00003>.